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Implicit Force Control for an Industrial Robot based on Stiffness Estimation and Compensation during Motion

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Abstract— **Although force control algorithms have been studied for three decades, this technology is not largely exploited in industry yet. The present paper proposes a position-based adaptive force control strategy, that relies on a novel method for the on line estimation of the environment stiffness. The control design is targeted to industrial controller structures and it is theoretically proven to be robust to time varying estimation errors of the environment stiffness and joint friction disturbances. The estimation algorithm succeeds in identifying the environment stiffness even in presence of geometrical irregularities of the contact surface during motion. The identification and control approaches are experimentally validated on an industrial robot equipped with a force sensor.**

I. INTRODUCTION

The control of the contact force between a robot and the environment is highly desirable for a variety of applications that require physical interaction, as fettling and finishing operations or assembly. In the last three decades, interaction control methods have been intensively studied [1]. Nevertheless, force control technology is currently only partially exploited. In order to achieve an effective technology transfer from research labs to industry, force control algorithms should be compatible with industrial controllers and robust to the compliance of the system. In fact, as industrial robots are becoming lighter and lighter, control algorithms should be designed considering the compliance of the overall system, composed by robot links, joints and environment.

Several compliance models have been proposed in the literature. The simplest models take into account just joints compliance [2]–[4], neglecting link flexibility [5], [6] and environment elasticity [7]. On the basis of the specific compliance model adopted, different control strategies have been proposed in the literature. In [7], [8] robot compliance is identified offline and included in the control algorithm. When no information on stiffness parameters is available, a robust control can be applied as in [9] or compliance parameters can be estimated and compensated with an adaptive approach [10]–[12]. In other works, estimation and control algorithms are strictly interdependent [13]–[16]. When the total stiffness of the system is known and compensated, higher force tracking performance can be achieved together with increased stability robustness. Thus, the on-line estimation of the environment compliance is a critical operation for the effectiveness of force control algorithms.

Identification of compliance parameters can be addressed to simulation or detection and perception objectives, or it can be carried out for a control strategy. The methods developed from the former perspectives are more accurate and they derive from non linear and complex models [17]–[19], while control-inspired estimation algorithms are established on simpler linear models. Among the latter methods, several approaches have been proposed: RLS [20], [21], EKF [22]–[24], algorithms based on an active observer [16], or strictly linked to the control strategy as Model Reference Adaptive Control methods [13]–[15], [25]. When the stiffness of the overall system is sufficiently high, the unknown geometrical irregularities of the contact surface can be extremely critical. The approaches that neglect this effect are suitable for applications characterized by systems with low stiffness and a good knowledge of environment geometry [13], [16], [20], [21], [25]. In other methods, it is assumed that the surface shape is a priori known and the compliance and the geometry parameters are jointly estimated, either by identifying a single offset parameter of the plane [14], [23] or a complex surface [22]. In an interesting approach, the stiffness parameters are estimated together with the normal and tangential directions of the contact surface and the friction coefficients [23]. It results in a quite massive system of equations that can be sensitive to numerical errors.

The present work discusses a novel approach for the estimation of the environment stiffness that is robust to geometrical irregularities of the contact surface, and an adaptive control algorithm independent from the elastic property of robot and environment. The estimation process is performed by adding a bandwidth-limited signal to the end effector motion and filtering the resulting contact force measure. The filtered signals are passed to an Exponentially Weighted Recursive Least Squares algorithm (EWRLS, [26]) to obtain the estimate of the environment stiffness. By identifying off line the elastic parameters of the robot joints and links, as in [7], the output of the on line estimation process turns out to be an accurate estimate of the environment stiffness, independent from robot configuration. The proposed approach does not require the estimation of parameters describing the environment geometry nor the knowledge of the initial deformation value.

The estimate is exploited in a simple adaptive position-based control algorithm, which can correctly assign a desired force response dynamics independently from robot configuration. The control strategy is theoretically proven to be robust to time varying error on stiffness estimate and to strong joint friction disturbances. The position-based design is suitable for the structure of industrial robot controllers, thus it could be directly exploited in technological applications.

The approach is extensively validated by experiments that show the accuracy of the estimate, both in static condition and during motion, and prove the effectiveness of the adaptive control algorithm.

The paper is organized as follows. In Section II the adopted elastic model is briefly reported, while in Section III the estimation algorithm is presented. The control strategy is

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reported in Section IV together with the stability discussion. The description of the experimental validation is presented in Section V and the conclusions complete the work.

II. ELASTIC MODEL

The system composed of an industrial robot in contact with the environment can be modeled as a series of equivalent springs at the end effector as in [7].
This approach assumes that, when a contact force is exchanged between environment and robot end effector, the elastic deformations of joints, links and environment can be reported to the contact point as equivalent end effector displacements. It relies on the hypothesis that deformations due to the contact force are sufficiently small to neglect the robot configuration changes.
The three equivalent compliances at the end effector can be computed as follows, where the purely elastic behavior is considered:

$$\begin{aligned}\boldsymbol{x}_e - \boldsymbol{x}_s &= -R_c\, K_E^{-1}\, R_c^T \boldsymbol{W}_e = -R_c\, C_E\, R_c^T\, \boldsymbol{W}_e,\\ \Delta\boldsymbol{x}_{e,J} &= J\left(\boldsymbol{q}\right)^{-1} K_M^{-1} J\left(\boldsymbol{q}\right)^T \boldsymbol{W}_e = C_J\, \boldsymbol{W}_e,\\ \Delta\boldsymbol{x}_{e,L} &= \sum_{i=1}^{6} A_{L_i}(\boldsymbol{q})^T\, K_{L_i}^{-1} A_{L_i}(\boldsymbol{q})\, \boldsymbol{W}_e = C_L\, \boldsymbol{W}_e\end{aligned}$$

where $\boldsymbol{x}_e \in \mathbb{R}^6$ is the actual end effector pose; $\boldsymbol{x}_s \in \mathbb{R}^6$ is the nominal undeformed pose on the environment surface; $\boldsymbol{W}_e \in \mathbb{R}^6$, $\boldsymbol{W}_e = \begin{bmatrix}\boldsymbol{F}_e^T & \boldsymbol{M}_e^T\end{bmatrix}^T$ is the contact wrench at the end effector; $\boldsymbol{q} \in \mathbb{R}^6$ are the link-side robot joint coordinates; $R_c \in \mathbb{R}^{6\times m}$ is a matrix whose columns are the m generalized directions elastically constrained by the environment; $J\left(\boldsymbol{q}\right) \in \mathbb{R}^{6\times 6}$ is the robot Jacobian; $A_{L_i}\left(\boldsymbol{q}\right) \in \mathbb{R}^{6\times 6}$ is the matrix that maps the twist on the end effector to the end of link i; $\Delta\boldsymbol{x}_{e,J}, \Delta\boldsymbol{x}_{e,L} \in \mathbb{R}^6$ are the end effector generalized displacements related to joint and link deformations; $K_E \in \mathbb{R}^{m\times m}, K_M, K_{L_i} \in \mathbb{R}^{6\times 6}$ are the structural stiffnesses of environment, joints and a generic link i, while $C_E \in \mathbb{R}^{m\times m}$, C_J, $C_L \in \mathbb{R}^{6\times 6}$ are the equivalent environment, joint and link compliances at the end effector. The complementary stiffness matrix for joint compliances has not been considered in the analysis for simplicity (see [27] for the definition).
Given $\boldsymbol{q}_m \in \mathbb{R}^6$ the motor-side robot joint coordinates, let's define $\boldsymbol{x}_e^{qm} \in \mathbb{R}^6$, $\boldsymbol{x}_e^{qm} = \boldsymbol{f}\left(\boldsymbol{q}_m\right)$ the pose of the end effector in the case of perfectly rigid robot. Thus the actual end effector pose $\boldsymbol{x}_e$ is given by:

$$\boldsymbol{x}_e = \boldsymbol{x}_e^{qm} + \Delta\boldsymbol{x}_{e,J} + \Delta\boldsymbol{x}_{e,L}.$$

Finally, the motor-side joint displacement $\tilde{\boldsymbol{q}}_m$ that corresponds to the contact wrench $\boldsymbol{W}_e$ can be computed as follows:

$$\tilde{\boldsymbol{q}}_m = -J^{-1}\left(R_c\, C_E\, R_c^T + C_J + C_L\right)\boldsymbol{W}_e, \qquad (1)$$

where the hypothesis of small displacements has been enforced. In the next Section, a discussion on the estimation of the compliance matrices is presented.

III. COMPLIANCE ESTIMATION

The compliance matrices of robot joints and links should be estimated off line with an ad hoc procedure, while the environment stiffness can be estimated either off line or on line. The adopted method for the estimation of joint and link compliances is described in [7]. A Least Square Minimization is exploited on the model regressors ϕ, containing all the stiffness parameters of joint and link matrices K_M and K_{L_i}: the torsional stiffness of each joint and, at most, four parameters for each link, modeled as a Bernoulli beam, representing the two bending stiffnesses along the principal axes of inertia of the section, the axial stiffness and the torsional stiffness. See [7] for further details.
In the following, the estimation of the environment compliance will be presented.

A. On-Line Estimation of Environment Compliance during motion

The estimation of environment stiffness, when it is supposed constant in every point of the contact surface, can be easily included in the previous off-line approach. Equivalently, it can be computed with a state of the art algorithm before the robot starts the task in contact with the surface. On the other hand, when the environment stiffness is unknown and cannot be supposed constant a priori, an on-line estimation is necessary. The on-line estimation while the robot is moving on the surface is particularly critical due to the possible geometrical irregularities of the surface. In fact, the end effector displacement along the direction of the applied force is due to the compliance of the environment and the geometry of the surface as well. If the surface has a small inclination along the direction of motion, the end effector should comply with the surface shape if the magnitude of contact force has to be kept constant. Thus, just a component of the displacement is proportionally related to force magnitude, while the other one is due to surface geometry.
State of the art algorithms for stiffness estimation can not discriminate between the two effects, while for a stiff environment even small inclinations of the surface can compromise the estimate. A different approach, robust to small errors on surface geometry, is here presented.
In this Section, environment compliance is considered just a linear spring in the normal direction, assuming a point contact.

1) Unbiased Estimator Computation: We define $\boldsymbol{p} = \begin{bmatrix}x_p & y_p & z_p\end{bmatrix}^T$, $\boldsymbol{p} \in \mathbb{R}^3$ the position of the end effector with reference to an inertial frame fixed with the environment. Then, we define in the same reference frame the contact force $\boldsymbol{F}_e \in \mathbb{R}^3$ and the contact surface $g\left(x,y,z\right) = 0$. We assume that the surface can be expressed in the following form: $s_z = s_z\left(x,y\right)$, at least in a neighborhood $\mathcal{B}_o$ of a given position $\boldsymbol{p}_o$.
It is assumed that the normal direction to the surface is $\boldsymbol{n}(x,y)$ while the estimation of the environment stiffness will be performed along a direction $\hat{\boldsymbol{n}}$, an estimate of the surface average normal. For simplicity, without any loss of generality, it will be assumed that $\hat{\boldsymbol{n}}$ is fixed and parallel to the z axis, and that $\boldsymbol{n}(x,y)$ has small deviations with respect to $\hat{\boldsymbol{n}}$. We can express the contact force measured in the direction $\hat{\boldsymbol{n}}$, by projecting onto $\hat{\boldsymbol{n}}$ the force due to the surface penetration of the end effector $\boldsymbol{p}$ with respect to the

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corresponding point on the surface $\boldsymbol{p}_s$:

$$F_e(t) = -K_e(t)\,\hat{\boldsymbol{n}}^T\left(\boldsymbol{n}(t)\,\boldsymbol{n}(t)^T\right)\left(\boldsymbol{p}(t) - \boldsymbol{p}_s(t)\right)$$

where $F_e = \hat{\boldsymbol{n}}^T\boldsymbol{F}_e$, $\boldsymbol{n} = \boldsymbol{n}(x_p(t), y_p(t))$, $\boldsymbol{p}_s = \begin{bmatrix} x_p(t) & y_p(t) & s_z(x_p(t), y_p(t)) \end{bmatrix}^T$ and $K_e(t) = K_e\big(x_p(t), y_p(t)\big)$. The end effector position $\boldsymbol{p} = \begin{bmatrix} x_p(t) & y_p(t) & z_p(t) \end{bmatrix}^T$ is just a function of time, while the actual environment compliance K_e, the surface normal $\boldsymbol{n}$ and the surface reference point $\boldsymbol{p}_s$ depend obviously on the current position of the end effector itself.

It is assumed that the end effector will not lose the contact with the surface. Clearly, as z axis is parallel to $\hat{\boldsymbol{n}}$ direction, it results:

$$\begin{aligned} \boldsymbol{p}(t) - \boldsymbol{p}_s(t) &= \hat{\boldsymbol{n}}\Big(z_p(t) - s_z\big(x_p(t), y_p(t)\big)\Big) \\ F_e(t) &= -K_e(t)\,N(t)\Big(z_p(t) - s_z\big(x_p(t), y_p(t)\big)\Big), \end{aligned} \quad (2)$$

where $N(t) = \hat{\boldsymbol{n}}^T\left(\boldsymbol{n}(t)\,\boldsymbol{n}(t)^T\right)\hat{\boldsymbol{n}}$, $N \in \mathbb{R}$, is just a projection operator.

First, let us study the effects of geometry and stiffness variation, together with the end effector displacement, on the measured force. To this end, let's formulate the problem in velocity form. Applying the time derivative to the contact force $F_e(t)$ in (2), we can write:

$$\frac{dF_e}{dt} = \nabla_{xy}F_e\,\boldsymbol{v}_{xy} + \frac{\partial F_e}{\partial t},$$

where $\nabla_{xy}(*) = \begin{bmatrix} \frac{\partial *}{\partial x} & \frac{\partial *}{\partial y} \end{bmatrix}$ and $\boldsymbol{v}_{xy} = \begin{bmatrix} \dot{x}_p & \dot{y}_p \end{bmatrix}^T$. Thus:

$$\begin{aligned} \frac{dF_e}{dt} =& (\nabla_{xy}K_e)\,N\,(s_z - z_p)\,\boldsymbol{v}_{xy} + K_e(\nabla_{xy}N)\,(s_z - z_p)\,\boldsymbol{v}_{xy} + \\ &+ K_eN\,(\nabla_{xy}s_z)\,\boldsymbol{v}_{xy} - K_eN\frac{\partial z_p}{\partial t}. \end{aligned}$$

The first term is due to the variation of stiffness on the surface, the second and third terms represent the effects of the geometrical variation of the contact surface on the force, while the last one is due to the variation of the end effector position along the z axis. When the robot has no tangential velocity, only the last term is different from zero.

It is useful to separate the parts depending on z_p and on $\frac{dz_p}{dt}$ from the other ones (the partial derivative $\frac{\partial z_p}{\partial t}$ can be substituted by the total derivative $\frac{dz_p}{dt}$):

$$\begin{aligned} \frac{dF_e}{dt} =& \Big(\big((\nabla_{xy}K_e)\,N + K_e\,(\nabla_{xy}N)\big)s_z + K_eN\,(\nabla_{xy}s_z)\Big)\boldsymbol{v}_{xy} + \\ &-\big((\nabla_{xy}K_e)\,N + K_e\,(\nabla_{xy}N)\big)\boldsymbol{v}_{xy}\,z_p - K_eN\frac{dz_p}{dt}\ . \end{aligned}$$

We then assume that all the time-varying terms are quasi-static compared with z_p and $\frac{dz_p}{dt}$. It means that the geometrical terms K_e, N, s_z are sufficiently smooth with respect to x and y coordinates, while the velocity $\boldsymbol{v}_{xy}$ is composed of low frequency components only. This assumption is consistent with the fact that sufficiently high frequency components will be added on purpose to z_p for the estimation.

We can now compute the Laplace transformation of the previous expression:

$$s\mathcal{L}[F_e](s) = \frac{A}{s} + B\mathcal{L}[z_p](s) + Cs\,\mathcal{L}[z_p](s), \quad (3)$$

where:

$$\begin{aligned} A &= \Big(\big((\nabla_{xy}K_e)\,N + K_e\,(\nabla_{xy}N)\big)s_z + K_eN\,(\nabla_{xy}s_z)\Big)\boldsymbol{v}_{xy} \\ B &= -\big((\nabla_{xy}K_e)\,N + K_e\,(\nabla_{xy}N)\big)\boldsymbol{v}_{xy},\ C = -K_eN. \end{aligned} \quad (4)$$

Elaborating on (3) we obtain:

$$\mathcal{L}[F_e](s) = \frac{A}{s^2} + \frac{B}{s}\mathcal{L}[z_p](s) + C\mathcal{L}[z_p](s), \quad (5)$$

which relates the Laplace transform $\mathcal{L}[z_p](s)$ of the end effector displacement along z axis to the Laplace transform of the contact force $\mathcal{L}[F_e](s)$, measured in the same direction. Let's define the end effector displacement $z_p(t)$ as composed of two components:

$$z_p(t) = z_p^N(t) + z_p^E(t),\ z_p^E(t) = A_E\sin(\omega_E t)$$

where $z_p^N(t)$ is the nominal end effector displacement and $z_p^E(t)$ is a periodic component added for estimation purpose. The $z_p^E(t)$ component will be added during motion in order to excite the high frequency force response, that depends on the environment compliance K_e without being affected by the robot motion $\boldsymbol{v}_{xy}$ (see coefficient C in (4)). Obviously, the magnitude A_E of the sinusoid should be selected as low as possible in order to have a negligible effect on robot task. Thus, for estimation purpose, an ideal band-pass filter $b(s)$ at frequency ω_E is applied to (5), in order to isolate the response to the added component:

$$\begin{aligned} &b_{\omega_E}(i\,\omega)\,\mathcal{L}[F_e](i\,\omega) = \mathcal{L}[F_e](i\,\omega_E), \quad \omega = \omega_E \\ &b_{\omega_E}(i\,\omega)\,\mathcal{L}[F_e](i\,\omega) = 0, \quad \omega \neq \omega_E. \end{aligned} \quad (6)$$

As the component $z_p^E(t)$ is a sinusoid at frequency ω_E, it can be assumed that $|Z_p(i\,\omega_E)| \gg 0$ where $Z_p(s)$ is the Laplace transforms of $z_p(t)$. Then:

$$\mathcal{L}[F_e](i\,\omega_E) = -\frac{A}{\omega_E^2} - \frac{iB}{\omega_E}Z_p(i\,\omega_E) + CZ_p(i\,\omega_E), \quad (7)$$

If ω_E is a sufficiently high frequency, the amplitudes of the first two terms are negligible with respect the third one, and (7) can be approximated and inserted into (6):

$$\begin{aligned} &b_{\omega_E}(i\,\omega)\,\mathcal{L}[F_e](i\,\omega) \approx -K_eN\,Z_p(i\,\omega_E), \quad \omega = \omega_E \\ &b_{\omega_E}(i\,\omega)\,\mathcal{L}[F_e](i\,\omega) = 0, \quad \omega \neq \omega_E. \end{aligned}$$

where the definition of $C = -K_eN$ has been used. If the value of N is close to one, i.e. a sufficiently good estimate of the surface normal is available, an explicit relation for the calculation of the environment stiffness K_e is given. Still, if the deviations of N have zero mean, their effects are filtered. Therefore an estimate of the environment stiffness is given by the following expression:

$$\hat{K}_e = \frac{F_e^{\omega_E}}{Z_p^{\omega_E}}, \quad (8)$$

where $F_e^{\omega_E}$ and $Z_p^{\omega_E}$ are the magnitude of the harmonic components of signals $F_e(t)$ and $z_p(t)$, respectively, at frequency $\omega = \omega_E$.

As far as the actual computation of the end effector displacement $z_p(t)$ is concerned, if no information about robot

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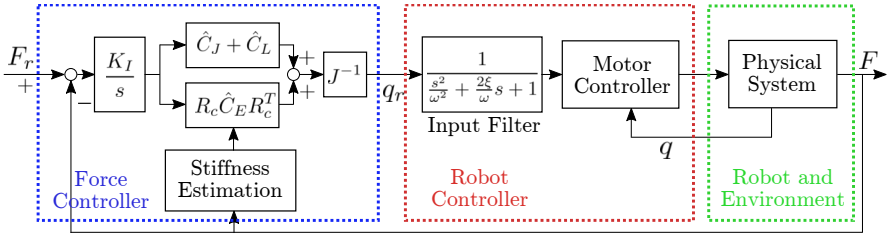


Fig. 1: Force control structure synthesized for an industrial robot. The feedback of q on the force controller blocks has not been reported for clarity.

compliance is available, the kinematic relation derived by the rigid model can be applied:

$$z_p = f_z\left(\boldsymbol{q}_m\right), \tag{9}$$

where f_z is the direct kinematics of the robot computed in z direction. In this case, the estimate $\hat{K}_e$ will include all the compliances of the elastic chain, thus robot joint and link compliances, together with the environment compliance. Notice that considering the total compliance does not influence the correctness of the previous result, under the hypothesis that $z_p^E\left(t\right)$ component causes sufficiently small changes of robot configuration.

On the other hand, when an off-line estimation of robot compliances is available, the value $z_p\left(t\right)$ can be computed by combining robot kinematics and static robot deformation:

$$z_p = f_z\left(\boldsymbol{q}_m\right) - \hat{\boldsymbol{r}}^T\left(C_J + C_L\right)\boldsymbol{W}_e,$$

or, for small displacements as in the formulation (1):

$$z_p = z_{p,0} + J_z\tilde{\boldsymbol{q}}_m - \hat{\boldsymbol{r}}^T\left(C_J + C_L\right)\boldsymbol{W}_e, \tag{10}$$

where $\hat{\boldsymbol{r}}$ is the estimated generalized direction constrained by the environment, whose first 3 components are $\hat{\boldsymbol{n}}$. For the defined reference system and a point contact, $\hat{\boldsymbol{r}} = \begin{bmatrix}0 & 0 & 1 & 0 & 0 & 0\end{bmatrix}^T$. With this approach, the value $\hat{K}_e$ represents an estimation just of the environment stiffness.

2) On-line Estimation Algorithm: As the expression in (8) is an unbiased estimate of the desired stiffness, the process can be implemented on-line as follows.

An on-line band-pass filter is applied to the force and end effector displacement signals $F_e(t)$ and $z_p(t)$ (computed as in (9) or in (10)) at frequency ω_E, obtaining the filtered signals $\bar{F}_e(t) \approx \mathcal{L}^{-1}\left[\mathcal{L}\left[F_e\right]\left(i\,\omega_E\right)\right]$ and $\bar{z}_p(t) \approx \mathcal{L}^{-1}\left[Z_p\left(i\,\omega_E\right)\right]$. The conditioned signals are passed as inputs to the well known EWRLS (see [26]) algorithm to obtain an on-line estimate $\hat{K}_e$ of the environment compliance $K_e(t)$ (or the total compliance when formulation (9) is used). When the robot is not moving, the band-pass filter is not necessary, and the original signals can be used as inputs. This approach, with the robot at rest, will be applied in Section V-A for validation purposes.

IV. CONTROL APPROACH

In this Section we propose an adaptive control law that compensates the entire elastic chain. A discussion about the stability of the system is then developed.

A. Control Structure

The control strategy is synthesized for a typical industrial robot, whose controller takes as input the joint references and computes the motor torques. Thus, the force control loop here proposed is closed externally to the industrial controller, taking as input the end effector force error and giving as output the joint references. The control structure is shown in Fig. 1 and it derives from the compliance model presented in Section II. The joint motor reference coordinates $\boldsymbol{q}_m^*$ are computed as follows:

$$\boldsymbol{q}_m^* = -J^{-1}\left(R_c\,\hat{C}_E\,R_c^T + \hat{C}_J + \hat{C}_L\right)\frac{K_I}{s}\left(\boldsymbol{W}_e^* - \boldsymbol{W}_e\right), \tag{11}$$

where $\boldsymbol{W}_e^*$ is the reference wrench to be tracked, K_I is the integral gain of the force control loop, and $\hat{C}_E$, $\hat{C}_J$, $\hat{C}_L$ are the estimated compliance matrices of environment, joints and links, respectively. In the following, the generic expression in (11) will be applied to the case where only a scalar force F is controlled with reference F^*, where $\boldsymbol{W}_e^* = \boldsymbol{r}\,F^*$, $\boldsymbol{W}_e = \boldsymbol{r}\,F$ and $R_c = \boldsymbol{r}$, $\boldsymbol{r} \in \mathbb{R}^6$. Thus, the outcome of the estimation algorithms presented in the previous Section is exploited in the control strategy, such that the force tracking performance is made independent from robot configuration and environment compliance. Remind that the estimation matrices of joint and link compliances $\hat{C}_J$ and $\hat{C}_L$ are computed on-line as a function of the robot configuration, but the parameters of the matrices are obtained off-line. Instead, the environment compliance $\hat{C}_E$ can be estimated either off-line or on-line. In the latter case the control adapts to changes of the environment compliance.

When no information on $\hat{C}_J$ and $\hat{C}_L$ are previously available, e.g. the off-line parameter identification has not been performed, the control algorithm can be rewritten as follows:

$$\boldsymbol{q}_m^* = -J^{-1}\hat{C}_T\frac{K_I}{s}\left(F_e^* - F_e\right),$$

where the equivalent total stiffness $\hat{C}_T$ has to be estimated on-line as shown in the previous Section.

B. Stability Discussion

The most relevant aspects of the stability analysis are related to the choice of the control gain K_I and the estimation error of the compliance matrices. As the environment elasticity is supposed to change along time, the equivalent estimation error on the total gain will be considered time varying. The system will be analyzed taking into account a single degree of freedom scheme (see Fig. 2a). The model neglects the robot dynamics because, at the imposed

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frequencies of the force control loop, the inner robot joint control can be regarded as ideal and the robot shows quasi-static behavior. This would not be suitable for infinitely rigid robots with stiff environment, but it is not the considered case.

Generally, industrial robot controllers apply a low-pass filter action on the joint reference signal to avoid the excitation of the mechanical resonance frequencies of the robot. Thus, a second order low-pass filter is included in the scheme.

The system is assumed to be independent from robot joint configuration, because the total estimated compliance matrix $\hat{C}_T$ and the total real stiffness matrix K_T are slowly varying with respect to joint coordinates $\boldsymbol{q}$ while the robot is moving, and force control loop imposes small displacements. Thus, Jacobian matrices can be considered constant in the analysis and the dependence on $\boldsymbol{q}$ will be neglected.

A delay has been introduced in the scheme (Fig. 2a). It models the joint static friction effect, which can be relevant in particular conditions when the force control loop is active, for stiff robot and environment and small force error. In particular, when the joint motion is inverted, a delay t_d occurs in joint position tracking, corresponding to the time taken by the integral action of the joint controller to overcome the static friction torque τ_F: $t_d \approx \frac{\tau_F}{K_{I,J}(q^* - q)}$.

The proposed scheme can be reordered as in Fig. 2b where the elements in Fig. 2a have been grouped in a linear function $L(s)$ and a time-varying gain $\psi(t)$ as follows:

$$\psi(t) = K_I \hat{C}_T(t) K_T(t),$$
$$L(s) = \frac{1}{s} \frac{\omega^2}{s^2 + 2\xi\omega s + \omega^2} H_{t_d}(s).$$

The gain $\psi(t)$ is the product of the integral force controller gain, the estimated total compliance and the real total stiffness. It depends on the variation of the real total stiffness K_T and on the accuracy of stiffness estimation $\hat{K}_T = \hat{C}_T^{-1}$. The linear part $L(s)$ is a third order transfer function with unitary gain, composed of the integral action of the force controller, the second order low-pass filter of the robot controller and possibly a linear Padé approximant $H_{t_d}(s)$ of the delay t_d.

The stability of the system can be analyzed with the circle criterion. When $\psi(t)$ takes values included in the interval $[k_1, k_2]$ $k_1, k_2 \in \mathbb{R}$, the system is absolutely stable if the Nyquist plot of $L(s)$ does not intersect the circle whose diameter is the segment between the points $\left\{\frac{1}{k_1}, 0\right\}$ and $\left\{-\frac{1}{k_2}, 0\right\}$ in the complex plane. In Fig. 3, the Nyquist plots of $L(s)$ without and with a Padé approximant delay have been drawn (blue solid and blue dashed lines respectively). The considered delay has been upper bounded to $t_d = 0.2s$ and described by a third order approximant. The low-pass filter takes the same parameters of the filter implemented in the experimental setup that will be described in the next Section ($\omega = 12$ Hz, $\xi = 0.99$). Then, assuming a value of K_I equal to 2 (such that the bandwidth of the controlled system with $\hat{C}_T(t)\, K_T = 1$ would be exactly 2 rad/s), two circles with diameter $[-\frac{1}{K_I \sigma_{\min,i}}, -\frac{1}{K_I \sigma_{\max,i}}]$ have to be drawn (the i index is equal to 1 or 2 for the systems without and with delay respectively). The coefficients $\sigma_{\max,i} = \max_t \hat{C}_T(t) K_T(t)$ and $\sigma_{\min,i} = \min_t \hat{C}_T(t) K_T(t)$ represent the maximum and minimum estimation error rates before the system becomes unstable.

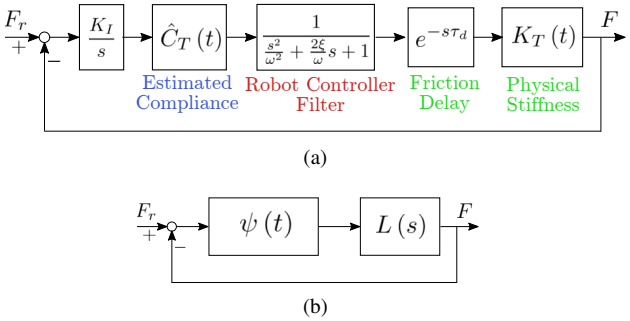


Fig. 2: Simplified schemes for stability discussion.

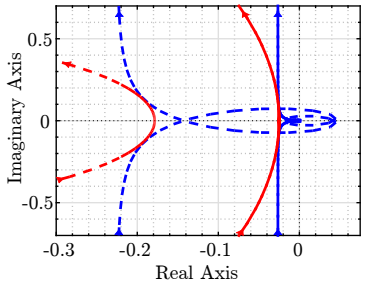


Fig. 3: Nyquist plot for the Circle Criterion. The blue lines are the $L(s)$ function without (solid) and with (dashed) delay, while red lines are corresponding circles for the $\psi(t)$ parameter at stability limits.

The values of $\sigma_{\max,i}$ and $\sigma_{\min,i}$ have been found graphically, taking $\sigma_{\min,i} = \frac{1}{\sigma_{\max,i}}$ (this is an arbitrary choice to have the same admissible error ratio for over and under estimation) and increasing $\sigma_{\max,i}$, starting from 1, until the circle would become tangent to $L(s)$. The two tangent circles are obtained when $\sigma_{\max,1} = 20$ for the system without delay (red solid line), and $\sigma_{\max,2} = 2.8$ for the system with delay (red dashed line). Thus, when friction does not affect relevantly the dynamics, the system is stable as long as the estimated stiffness varies arbitrarily between 20 times and $\frac{1}{20}$ the real stiffness. Notice that a lower integral gain K_I allows bigger uncertainty on the estimation error, as the limits of the circles are related to the product of the two quantities.

V. EXPERIMENTS

In this Section, the experimental validation of the proposed approaches is presented. The experimental setup consists of a six degrees of freedom robot COMAU Smart Six, 6 Kg payload industrial manipulator, equipped with the open version of the COMAU C4G controller. An ATI 6-axes wrist force/torque sensor is fitted to the arm end effector.

The results are presented incrementally. First, the off-line static estimation and compensation of environment stiffness is validated. Secondly, the on-line static estimation and compensation is presented. Finally, the on-line estimation and compensation of environment stiffness while the robot is moving is validated and discussed.

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The estimate of joint and link stiffness parameters is described in [7], and in the following the elastic parameters of the robot are supposed to be known.

A. Static Estimation and Compensation of Environment Compliance

The estimation of environment stiffness in static condition, i.e. the robot is not moving tangentially to the surface, is validated both with the on-line and off-line algorithms and applied in the control structure.

1) Off-Line Estimation and Control: The off-line estimation is performed with a simple Least Square Estimation (LSE) algorithm. Three experiments on different materials, rubber, wood and steel, are carried out. The robot end effector is put in contact with the given material, and then a force is exerted by the robot. The overall time history of the end effector displacements $\Delta \boldsymbol{z}$ and the corresponding measured forces $\Delta \boldsymbol{F}_z$ along the normal axis is exploited in the LSE method: $\hat{K}_E = -(\Delta \boldsymbol{z}^T \Delta \boldsymbol{z})^{-1} \Delta \boldsymbol{z}^T \Delta \boldsymbol{F}_z$.
As the joint and link compliances are available, the formulation in (10) is used to compute the end effector displacement $\Delta \boldsymbol{z}$. The obtained values for the three materials are reported in Table I, together with the mean square error of the given estimate. The obtained values are embedded

	Rubber	Wood	Steel
$\hat{K}_E$ [N/m]	$6.88 \cdot 10^4$	$2.87 \cdot 10^5$	$1.11 \cdot 10^6$
MSE [%]	5.2	8.62	12.4

TABLE I: Estimated Stiffness for three different materials by means of LSE algorithm.

in the control structure presented in (11) and a force control experiment is performed tracking a step reference. The force responses obtained in the three experiments are presented in Fig. 4. The color dashed lines correspond to

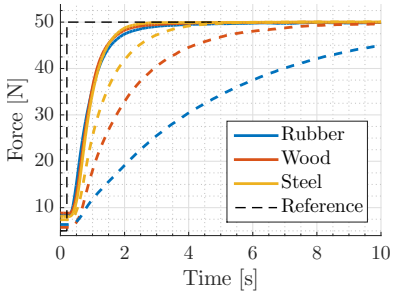


Fig. 4: Control with off-line computed environment stiffness.

the experiments in which the environment compliance is considered null. Obviously, the higher the real environment compliance is, the higher the force response time is. Solid lines correspond to experiments in which the stiffness values obtained off-line are used in the control algorithm. The behavior of the three experiments is indistinguishable, and it coincides with the theoretical desired response with time constant $0.5\,\mathrm{s}$, corresponding to the fixed gain $K_I = 2\,\mathrm{s}^{-1}$. This is a proof that all the compliances in the loop have been compensated.

2) On-line Estimation and Control: In order to perform an on-line estimation, the EWRLS algorithm is used as presented in Section III B. As the robot is not moving, the $z_p^E(t)$ component is not added.
In order to validate the method, we perform a contact experiment using commercial springs of known stiffness. Six experiments are performed in two different positions, in order to test the independence of the results from the robot configuration. The results are shown in Fig. 5, where the stiffness estimates for the two configurations (circle and square markers) are compared with the known stiffness of the springs, regarded as ground truth.

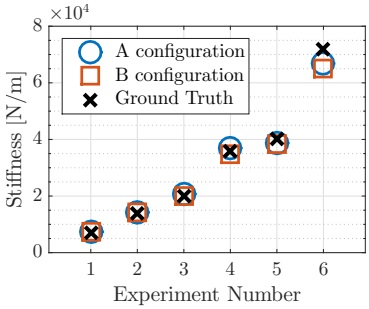


Fig. 5: Environment stiffness estimated obtained with the EWRLS algorithm. Two different configuration experiments (A and B) are compared with the known stiffness values (x).

The estimate turns out to be independent from the configuration and the average of the percentage absolute error is 4.1%. Thus the algorithm converges to the real value with a reasonable accuracy.
In order to test the performance of the control algorithm, while the environment stiffness is simultaneously estimated, a force control experiment is performed. Again, the environment stiffness is estimated on-line with EWRLS, and the control algorithm uses the current stiffness estimate value. The robot is not moving tangentially on the environment surface. The gain K_I is unitary, the forgetting factor of EWRLS is $\mu = 0.995$, and the estimate stiffness is initialized to be 5 times the real one. The experimental results are shown in Fig. 6. The blue line corresponds to the force behavior that

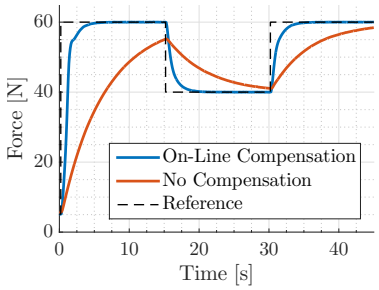


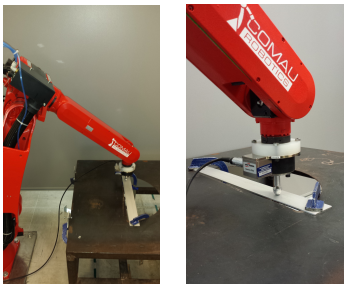
Fig. 6: Force control experiment with on-line estimation and compensation. The result is compared with an experiment without on-line compensation.

occurs when the control uses the current stiffness estimate, while the red line corresponds to an experiment in which the environment compliance estimate is imposed to be null. Notice that in 2 seconds the estimate converges to the final value.

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B. Estimation and Compensation of Environment Compliance during Motion

The next experiment aims at demonstrating the effectiveness and the theoretical validity of the proposed approach for the estimation and compensation of environment compliance, while the robot is moving tangentially on the contact surface. The end effector is put in contact with an aluminum beam, constrained on the extremities to a fixed support (see Fig. 7). A tangential velocity along the beam is imposed to the end



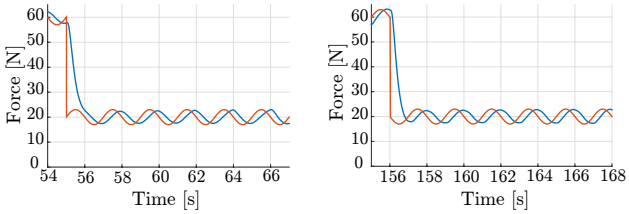
(a) (b)

Fig. 7: Force control experiment in contact with an aluminum beam. The equivalent environment stiffness is changing while the robot is moving along the beam itself.

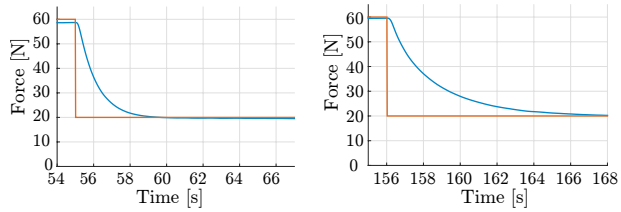
effector, while the contact force in z direction is controlled in order to track a given reference. During the motion, the perceived stiffness by the robot is dramatically changing, with the minimum in the middle of the beam.

The proposed approach is used to estimate and compensate the environment stiffness. Thus, a sinusoidal signal is added to the force reference, with amplitude $3\,\mathrm{N}$ and frequency $f_E = 0.5\,\mathrm{Hz}$. This is equivalent to imposing it to the end effector displacement. The filter is obviously set at $\omega_E = 2\pi\, f_E$. A FIR filter of order 100 is chosen with undersampling of the signals at $10\,\mathrm{Hz}$. The filter order is relevantly affecting the performance, because the higher the order of the filter is, the higher the delay in the estimation convergence is. On the other hand, the lower the chosen f_E is, the higher the filter order should be, and this frequency is upper limited by the force controller gain K_I (to correctly track the imposed sinusoidal signal). In order to obtain the best tracking of the reference force and to compare the experimental results with the ideal theoretical behavior, a low tangential velocity is imposed ($1\,\mathrm{mm/s}$), together with a gain $K_I = 5\,\mathrm{s}^{-1}$. A more efficient band-pass filter implementation would allow to increase the tangential velocity.

The imposed force reference and the actual force measured during the experiment are reported in Fig. 8. Notice that the force control with adaptive stiffness (plots 8a and 8b) is compared to the force control approach where a fixed value of stiffness estimated off-line is used (plots 8c and 8d). The plots on the left hand side (8a and 8c) represent a part of the two experiments corresponding to a position on the beam characterized by high stiffness, while the plots on the right hand side (8b and 8d) correspond to a position near the middle of the beam, thus with low stiffness. The estimated stiffness in the different time instants is shown in Fig. 9. Notice that the average estimated stiffness corresponding to time instants of 8a and 8c ($1.7\,10^5\,\mathrm{N/m}$) is about 10 times



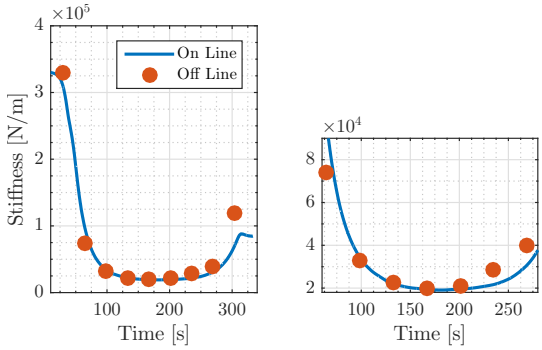
(a) On-line Estimate (Point A) (b) On-line Estimate (Point B)



(c) Off-line Estimate (Point A) (d) Off-line Estimate (Point B)

Fig. 8: Comparison between force tracking experiment with the adaptive stiffness approach (a and b) and force tracking with a fixed environment stiffness estimate (c and d) for two different positions on the beam (point A and B). Force references are in red while measures in blue.

the average estimated stiffness corresponding to time instants of 8b and 8d ($2.0\,10^4\,\mathrm{N/m}$). As the environment stiffness



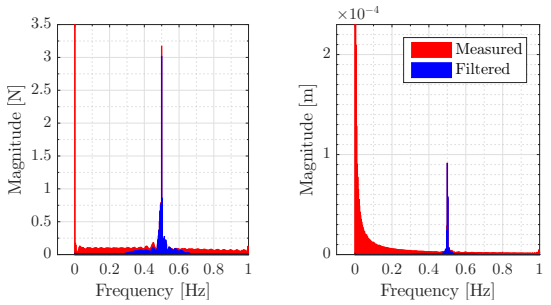
(a) Complete Experiment (b) Detail

Fig. 9: Environment stiffness on-line estimation along the beam during the experiment (blue line). Results are compared with the off-line static estimation (circles). In the plot (b) a detail of the plot (a) is reported. Notice that even small stiffness variations are detected by the algorithm.

is correctly estimated during motion, the force control is able to track the reference with the same performance along the experiment (plots 8a and 8b), while using the off-line estimated stiffness the performance degrades in points of the beam with smaller stiffness (plot d compared with plot c).

The spectra of force and end effector z displacement of the experiment are reported in Fig. 10. The imposed sinusoidal component is correctly filtered on both spectra. Notice that the spectrum of the displacement has the expected shape, derived theoretically. In fact, inverting (3) and observing that the modulus of $\mathcal{L}\left[F_e\right](i\omega)$ is negligible for $\omega > 0.02\pi\,\mathrm{rad/s}$, the modulus of the spectrum of the end effector z displace-

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(a) Force

(b) End Effector Displacement.

Fig. 10: Force and displacement spectrum during the experiment with imposed sinusoidal signal. The highest picks are out of scale, and they are $43.75\,\mathrm{N}$ for the force and $1.7\,10^{-3}\,\mathrm{m}$ for the displacement.

ment has the following form:

$$\mathcal{L}\left[z_p\right](i\omega) = \left|\frac{A}{i\omega\left(B + Ci\omega\right)}\right|, \quad \omega > 0.02\pi$$

which is perfectly compatible with the obtained experimental behavior. As a comparison, the same experiment was performed applying the EWRLS algorithm, without preconditioning the data, neither adding the sinusoidal component (in this case the force control loop employs an off line estimated value of environment stiffness for safety). The stiffness estimate turned out to be unstable due to the relatively large displacements (compared to the elastic deformation) needed to follow the surface geometry and keep the force reference.

VI. CONCLUSIONS

An adaptive implicit force control based on an on line estimation algorithm of environment stiffness is presented. The estimation algorithm is proved to be robust to geometrical irregularities of the surface and it does not require the estimation of geometrical parameters. The experiments validate the approach using as ground truth the nominal stiffness value of commercial springs for the static estimation, while the static measure itself is taken as ground truth for the estimation during motion. The control algorithm is adaptive in the environment stiffness parameter and it compensates the overall elastic chain that includes even robot joints and links. A theoretical proof of its robustness to estimation errors and joint friction disturbances is provided, and the experimental validation shows its effectiveness in tracking the force reference with the desired dynamics.

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